

Difference between non-tail and tail recursive-

Non-tail recursion can work after a recursive call while a tail recursion cannot. When it comes to stack usage, tail recursion is constant whereas non-tail recursion has n growth. The rate of performance for the fibonacci is $O(2^n)$ for non-tail recursion and $O(n)$ for tail recursion. Tail recursion has a very low risk of stack overflow while non-tail recursion is high.

Preferred-

Tail recursion is preferred because of its compiler optimization, stack overflow prevention, its capability to reuse stack frames, and its better compiler optimization.